

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	G.Hima Bindu
Register Number	192511377

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Hima Bindu*

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

Introduction :

Virtualization is a fundamental technology in cloud computing that allows multiple virtual machines (VMs) to run on a single physical server using a hypervisor. Hypervisors are of two types: **Type-1 (Bare-Metal)** and **Type-2 (Hosted)**. The choice of hypervisor affects performance, security, and resource management. Comparing these two types helps determine the best option for hosting mission-critical applications.

Comparative Analysis:

Dimension	Type-1 (Bare-Metal) Hypervisors (e.g., VMware ESXi, KVM)	Type-2 (Hosted) Hypervisors (e.g., VirtualBox, VMware Workstation)
Architecture	Installs directly on the bare physical hardware. Interacts directly with CPU, RAM, and I/O.	Installs as an application layer on top of a Host OS (e.g., Linux/Windows).
Performance	Near-native performance. Minimal overhead because execution skips the intermediate Host OS scheduler.	Higher latency & lower throughput. Every I/O request and CPU context switch travels through the Host OS system call stack.
Security	Small attack surface. Runs a minimal, hardened microkernel with dedicated driver models, reducing exploitable flaws.	Large attack surface. Inherits all vulnerabilities, background processes, and unpatched flaws of the underlying Host OS.
Manageability	Centralized enterprise orchestration (vCenter,	Decentralized management. Designed primarily for single-user

	OpenStack). Supports live migration (vMotion), dynamic resource allocation, and fault tolerance.	dev/test setups; lacks native multi-node clustering and live migration engines.
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Final Justification

For mission-critical applications (e.g., ERP systems, transactional banking databases),

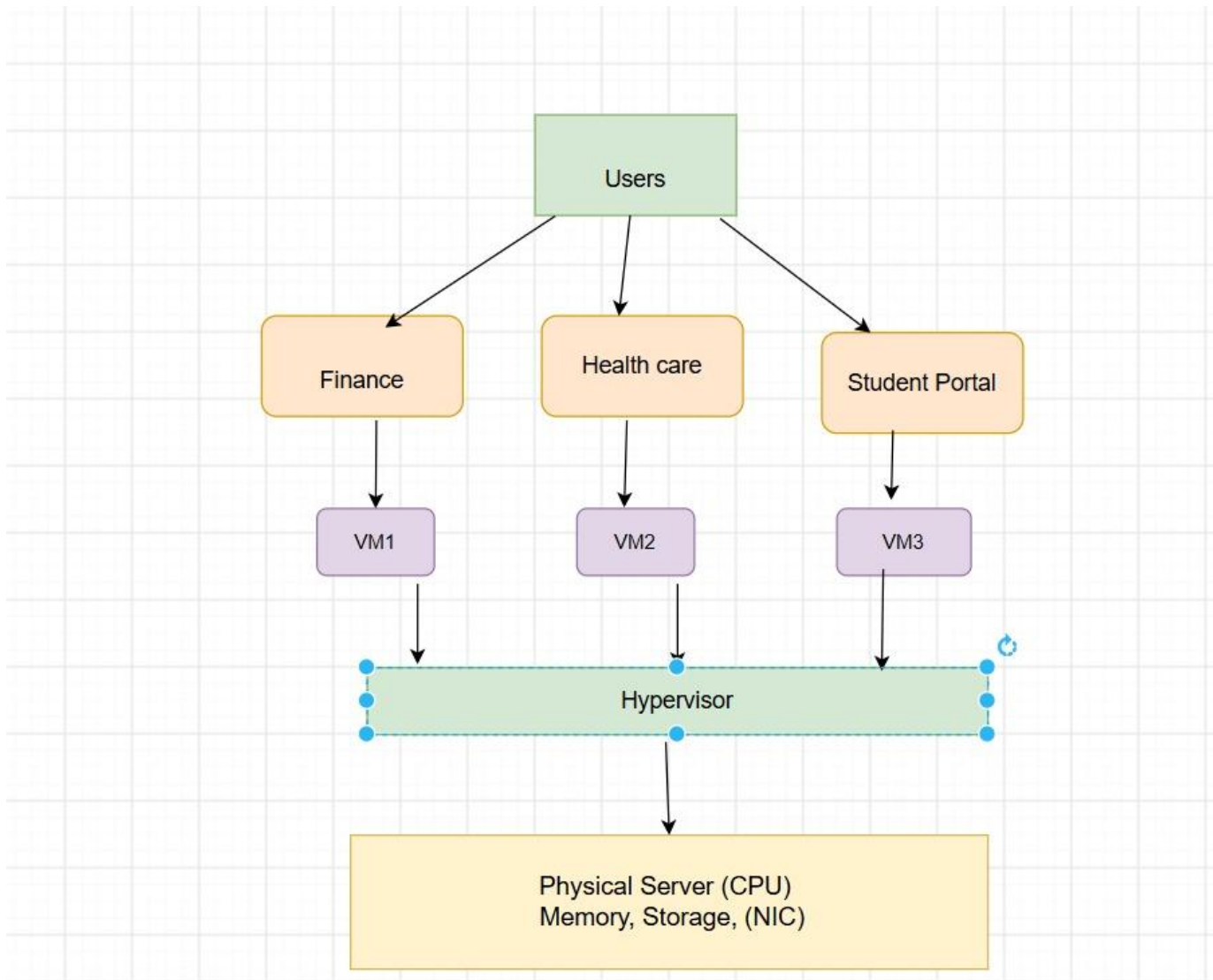
Type-1 hypervisors are the definitive choice.

1. **Uptime & Reliability:** They avoid single points of failure inherent to general-purpose host OS crashes or reboot cycles caused by background OS updates.
2. **Predictable I/O:** Direct hardware access ensures steady IOPS and ultra-low latency necessary for real-time transactions.
3. **Hardware Isolation:** They enforce hardware-enforced CPU/memory rings via Intel VT-x or AMD-V to guarantee true resource availability during usage spikes.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Introduction

Cloud providers often host multiple applications on the same physical server to improve resource utilization and reduce costs. In a multi-tenant environment, sensitive workloads such as finance, healthcare, and student portals must remain isolated to protect confidential data and ensure security. Virtualization architecture provides isolation by creating separate Virtual Machines (VMs) with dedicated virtual resources. This prevents one workload from interfering with another while maintaining efficient use of the underlying hardware.



How Virtualization Supports Isolation

1. CPU Isolation

- Each VM is allocated virtual CPUs (vCPUs).
- The hypervisor schedules CPU time separately for each VM.

Benefits:

- Prevents one workload from consuming all CPU resources.
- Ensures fair resource allocation.

2. Memory Isolation

- Every VM has its own dedicated virtual memory.
- One VM cannot access the memory of another VM.

Benefits:

- Prevents unauthorized access to sensitive data.
- Improves confidentiality and system stability.

3. Storage Isolation

- Each VM uses a separate virtual disk.
- Data is stored independently for every workload.

Benefits:

- Finance data remains separate from healthcare and student data.
- Prevents accidental or unauthorized data access.

4. Network Isolation

- Virtual switches (vSwitches), VLANs, or Software-Defined Networking (SDN) separate network traffic.

Benefits:

- Secure communication between applications.
- Prevents unauthorized network access between workloads.

5. Operating System Isolation

- Each VM runs its own operating system independently.

Benefits:

- OS failures or crashes in one VM do not affect others.
- Applications remain isolated and secure.

6. Security Policy Isolation

- Different security policies and firewall rules can be applied to each VM.

Examples:

- Finance VM: Strict financial security policies.
- Healthcare VM: Patient data protection policies.
- Student VM: Educational access policies.

Benefits:

- Better compliance with organizational and regulatory requirements.
- Stronger protection for sensitive workloads.

Major Attack Surfaces**1. Hypervisor Attack**

A hypervisor attack occurs when an attacker exploits vulnerabilities in the hypervisor.

Impact:

- Gain control over multiple VMs.
- Access confidential finance, healthcare, or student data.
- Disrupt cloud services.

Prevention:

- Keep the hypervisor updated.
- Apply security patches regularly.
- Use secure boot and access controls.

2. VM Escape Attack

VM Escape is a security attack where malicious software running inside one VM escapes to the hypervisor or another VM.

Impact:

- Cross-VM attacks.
- Data theft.
- Unauthorized access to other workloads.

Prevention:

- Strong VM isolation.
- Regular hypervisor updates.
- Least-privilege access controls.
- Continuous security monitoring.

Advantages of Virtualization Isolation

- Protects sensitive data between workloads.
- Enables secure multi-tenancy.
- Improves resource utilization.
- Prevents interference between applications.
- Supports compliance with security regulations.
- Enhances availability and reliability.

Conclusion

Virtualization architecture enables multiple workloads such as finance, healthcare, and student portals to run securely on the same physical hardware by providing CPU, memory, storage, network, operating system, and security policy isolation. However, the virtualization layer introduces security risks such as **hypervisor attacks** and VM escape attacks. These risks can be minimized through regular patching, strong access controls, secure configuration, and continuous monitoring, ensuring a secure and reliable cloud environment.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

VM host utilization snapshot and recommend corrective actions:

Diagnosis Matrix:

Metric	Measured Value	Threshold / Baseline	Status
CPU Utilization	92%	< 80%	Critical Overcommitment
Memory Utilization	88%	< 80%	High Pressure / Saturated
Storage I/O Latency	35 ms	< 10–15 ms	Severe Bottleneck
Avg. VM Boot Time	170 s	< 20–30 s	Severely Degrading

Root Cause Analysis

The primary system bottleneck is Storage I/O Bottleneck driven by Memory Swapping (Hypervisor Paging).

1. **The Cascade Effect:** With Memory at 88%, the hypervisor is forced to use balloon drivers, host swap files, or hypervisor swap disks to compensate for RAM overcommit.
2. **I/O Latency Explosion:** Writing unpaged memory pages to storage creates massive read/write queues. This pushes I/O latency to 35 ms (normal SSD SAN latency should be under 5 ms).
3. **Slow Boot Times:** The 170-second boot time is caused by VMs waiting on IOPS to read OS initialization files during startup (I/O starvation). High CPU usage (92%) further slows context switching under heavy software interrupt loads.

Recommended Corrective Actions

- **Immediate Actions (Mitigation):**

1. **Trigger Storage vMotion / Live Migration:** Immediately migrate 20–30% of non-critical VMs to an idle node in the cluster to relieve local CPU/RAM pressure.

2. **Set Memory Reservations:** Ensure critical production VMs have 100% memory reservations set to prevent them from being swapped to disk.

- **Long-Term Actions (Remediation):**

1. **Tune CPU/RAM Overcommit Ratios:** Lower physical-to-virtual vCPU overcommit ratios (v CPU :p CPU should drop from typical ~5:1 down to ~2:1).
2. **Storage Tiering & Flash Acceleration:** Upgrade storage backends to NVMe-over-Fabrics (NVMe-oF) or add local read/write SSD caches to absorb I/O queue spikes.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Traditional Datacenter

Characteristics:

- Physical servers
- Dedicated storage
- Hardware-based networking
- Manual configuration
- Slow provisioning

Problems:

- Poor scalability
- High operational cost
- Slow deployment

Software Defined Datacenter

All infrastructure is virtualized and managed through software.

Components:

- Compute virtualization
- Storage virtualization
- Network virtualization

Compute Virtualization

Traditional:

- One application per server.

SDDC:

- Multiple VMs share one server.

Benefits:

- Better utilization
- Rapid VM provisioning
- Easy migration

Storage Virtualization

Traditional:

- Dedicated storage arrays.

SDDC:

- Storage pooled into virtual storage.

Benefits:

- Higher flexibility
- Dynamic allocation
- Better disaster recovery

Network Virtualization

Traditional:

- Physical switches
- Manual configuration

SDDC:

- Virtual switches
- Software-defined networking (SDN)

Benefits:

- Fast network creation
- Better traffic management
- Improved security

Agility Comparison

Feature	Traditional	SDDC
Provisioning	Slow	Fast
Scalability	Limited	Highly scalable
Automation	Low	High
Resource Utilization	Moderate	High
Disaster Recovery	Difficult	Easier
Cost	Higher	Lower

Conclusion

SDDC improves agility by:

- Automating infrastructure management.
- Virtualizing compute, storage, and networking.

- Enabling rapid deployment, scalability, and efficient resource utilization.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Introduction

Multi-cloud federation enables organizations to use services from multiple cloud providers with a single user identity. However, different cloud providers use different identity management systems, authentication methods, and access policies. These differences can cause identity mismatches, resulting in login failures and security risks. A secure identity federation framework helps provide seamless and secure access across all cloud platforms.

Analysis of the Issue

1. Different Identity Formats

- Each cloud stores user identities differently.
- The same user may have different usernames in different clouds.

Impact: Login failures and duplicate user accounts.

2. Different Authentication Protocols

- Cloud providers may use SAML, OAuth 2.0, or OpenID Connect (OIDC).

Impact: Authentication requests may fail between providers.

3. Different Access Policies

- Each cloud has its own authorization rules.

Impact: Users may receive incorrect permissions or be denied access.

4. Privacy Risks

- Personal user information may be shared across clouds.

Impact: Data leakage and privacy concerns.

Secure Identity and Privacy Design

1. Central Identity Provider (IdP)

A trusted Identity Provider authenticates all users.

Benefits:

- Single user identity
- Centralized authentication
- Easy user management

2. Single Sign-On (SSO)

Users log in once and access all cloud services.

Benefits:

- Better user experience
- Reduces multiple passwords
- Simplifies login process

3. Standard Federation Protocols

Use standard protocols such as:

- SAML
- OAuth 2.0
- OpenID Connect (OIDC)

Benefits:

- Secure authentication
- Compatibility between cloud providers

4. Identity Broker

The Identity Broker connects the IdP with different cloud providers.

Functions:

- Token exchange

- Attribute mapping
- Identity translation

5. Role-Based Access Control (RBAC)

Users receive permissions based on their roles.

Example:

- Finance staff → Finance Cloud
- Doctors → Healthcare Cloud
- Students → Student Portal

Benefits:

- Prevents unauthorized access
- Implements least-privilege access

6. Multi-Factor Authentication (MFA)

Users verify their identity using more than one factor.

Example:

- Password + OTP

Benefits:

- Increases account security
- Protects against password theft

7. Encryption and Privacy

- Encrypt identity data during transmission and storage.
- Use pseudonyms instead of sharing personal information.

Benefits:

- Protects confidential data

- Enhances user privacy

Advantages

- Eliminates identity mismatches.
- Provides secure Single Sign-On (SSO).
- Improves interoperability between cloud providers.
- Protects user privacy with encryption and pseudonyms.
- Strengthens security using RBAC and MFA.
- Simplifies identity management.

Conclusion

Identity mismatches in a multi-cloud federation can lead to authentication failures and security issues. A secure solution includes a **Central Identity Provider (IdP), Identity Broker, SSO, SAML/OAuth/OIDC, RBAC, MFA**, and encryption. This design enables secure, seamless, and privacy-preserving access across multiple cloud providers.